

# Zhefeng Huang

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## EDUCATION

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|---------------------------------------------------------------------------------------------------------------------------------|--------------------------------|----------------|-----------------|
| GEORGIA INSTITUTE OF TECHNOLOGY                                                                                                 | Ph.D. in Robotics              | Atlanta, U.S.  | 08/2022-present |
| • Robotics Ph.D. Program, GPA: <b>4.00/4.00</b> ; Passed the Robotics Ph.D. Qualifying Examination                              |                                |                |                 |
| TSINGHUA UNIVERSITY                                                                                                             | B.S. in Mechanical Engineering | Beijing, China | 09/2018-06/2022 |
| • Major in Mechanical Engineering (Elite program), GPA: <b>3.83/4.00</b> ; Minor in Computer Application, GPA: <b>3.94/4.00</b> |                                |                |                 |

## RESEARCH EXPERIENCE

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| <b>World-Model-Augmented VLA for Long-Horizon Surgical Manipulation</b>                                                | Jun 2026 – Present  |
| • Developing world-model-augmented VLA policies for long-horizon surgical manipulation.                                |                     |
| • Learning future-predictive latent states from mixed-quality expert and non-expert interactions.                      |                     |
| <b>Flow-Matching World Model for Robot Manipulation Failure Detection</b>                                              | Feb 2026 – Jul 2026 |
| • Developed an action-conditioned flow-matching world model for failure detection using only successful training data. |                     |
| • Built simulation and real-world dVRK pipelines for data collection, imitation learning, and evaluation.              |                     |
| <b>Self-Reconfiguration Planning for Continuum Modular Robot System</b>                                                | Apr 2025 – Jan 2026 |
| • Developed a mathematical model of planar continuum modular robots using combinatorial topology.                      |                     |
| • Proposed HEART-MCTS for task planning and integrated Atlas-RRT* for motion planning.                                 |                     |
| • Developed a graph neural network (GNN)-based representation of modular robot morphology.                             |                     |
| <b>Mobile Nursing Robot in Intensive Care Unit</b>                                                                     | Apr 2024 – Oct 2024 |
| • Conducted a human-robot-interaction (HRI) study using a Stretch 2 robot deployed in an ICU ward.                     |                     |
| <b>Ultrasound-Guided Robotic Needle Insertion</b>                                                                      | Sep 2023 – Jan 2024 |
| • Developed a robotic needle insertion system using a Franka Research 3 arm with ultrasound guidance.                  |                     |
| <b>Development of an MR-Conditional Medical Robot</b>                                                                  | Jan 2023 – Sep 2023 |
| • Designed, developed and manufactured an MR-conditional robot prototype along with its mechatronics control system.   |                     |
| <b>Geometrical Modeling and Optimization of Concentric Tubes</b>                                                       | Sep 2022 – Jan 2023 |
| • Performed geometric modeling and optimization of concentric-tube robots for intracerebral hemorrhage evacuation.     |                     |

## SKILLS

**Robot Learning & AI:** Imitation Learning, World Models, Vision-Language-Action Models, Flow Matching  
**Robot Planning:** Task and Motion Planning, MCTS, Sampling-based Motion Planning, Robot Kinematics, Manipulation  
**Programming & Frameworks:** Python, C++, C, MATLAB; PyTorch, ROS 2, Isaac Sim, OpenCV, MuJoCo, Unity  
**Robot & Engineering Experience:** da Vinci Research Kit, Franka Research 3, Stretch 2; SolidWorks, ANSYS, COMSOL

## PUBLICATIONS

- **Huang, Zhefeng**, Yilin Cai, Ankit Patel, Mohammad Hajiha, Brendan Browne, and Yue Chen. "Failure Detection for Surgical Robot Imitation Policies via Flow-Matching World Modeling." *arXiv preprint arXiv:2607.27511*, 2026. Submitted to IEEE Robotics and Automation Letters (RA-L).
- Cai, Yilin\*, **Zhefeng Huang\***, Yifan Wang, Haokai Xu, and Yue Chen. "Evolutionary Diversification via Modular Compliance for Self-Reconfigurable Continuum Robots." *Science Advances* 12, eaeg9191, 2026. \*These authors contributed equally to this work.
- **Huang, Zhefeng**, Anthony L. Gunderman, Samuel E. Wilcox, Saikat Sengupta, Jay Shah, Aiming Lu, David Woodrum, and Yue Chen. "Body-mounted MR-conditional robot for minimally invasive liver intervention." *Annals of Biomedical Engineering* 52, no. 8 (2024): 2065-2075.
- **Huang, Zhefeng**, Hussain Alkhars, Anthony Gunderman, Dimitri Sigounas, Kevin Cleary, and Yue Chen. "Optimal concentric tube robot design for safe intracerebral hemorrhage removal." *Journal of mechanisms and robotics* 16, no. 8 (2024).
- Gunderman, Anthony L., Saikat Sengupta, **Zhefeng Huang**, Dimitri Sigounas, Chima Oluigbo, Isuru S. Godage, Kevin Cleary, and Yue Chen. "Towards mr-guided robotic intracerebral hemorrhage evacuation: Aiming device design and ex vivo ovine head trial." *IEEE Transactions on Medical Robotics and Bionics* (2024).
- Wilcox, Samuel, **Zhefeng Huang**, Jay Shah, Xiaofeng Yang, and Yue Chen. "Respiration-Induced Organ Motion Compensation: A Review." *Annals of Biomedical Engineering* 53, no. 2 (2025): 271-283.
- Sommer, Joseph, Anthony L. Gunderman, Saikat Sengupta, **Zhefeng Huang**, Dimitri Sigounas, Kevin Cleary, and Yue Chen. "Concentric Tube Robot-Based Intracerebral Hemorrhage Evacuation in ex vivo Sheep Head: A Comparative Study." In *2024 International Symposium on Medical Robotics (ISMR)*, pp. 1-7. IEEE, 2024.

## AWARDS & HONORS

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|-----------------------------------------------------------------------------------------|-----------|
| Outstanding Graduate, Tsinghua University                                               | 06/2022   |
| Three-Time Men's Group A High Jump Champion, 62nd–64th Ma John Cup, Tsinghua University | 2019-2021 |